SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: AI-Based Smart Water Leakage Detection System

Department of Electronics & Communication Engineering (ECE)

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SUBMITTED BY

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements, architecture, functional behavior, interfaces, constraints, and testing procedures for the **AI-Based Smart Water Leakage Detection and Automatic Protection System**. The system leverages IoT sensors, ESP32, MQTT communication, Machine Learning algorithms, and a Streamlit dashboard to detect water leakages in real time and automatically prevent water loss through intelligent control actions.

1.2 Scope of the System

The proposed system continuously monitors water pipelines using flow, pressure, and moisture sensors.

In-Scope Features

- Real-time sensor monitoring
- MQTT-based data communication
- AI-powered leakage prediction
- Streamlit web dashboard
- Automatic buzzer activation
- Automatic motor shutdown
- Automatic valve closure
- Historical data storage
- Leakage confidence prediction
- Report generation

Out-of-Scope Features

- Physical pipeline repair
- Municipal water supply management integration
- Commercial SCADA systems
- Water billing systems

Benefits

- Early leakage identification

- Water conservation
- Reduced maintenance costs
- Automatic safety actions
- Remote monitoring capability

1.3 Definitions and Acronyms

Acronym	Meaning
SRS	Software Requirement Specification
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
ESP32	Wi-Fi Enabled Microcontroller
UI	User Interface
DB	Database
API	Application Programming Interface
NFR	Non-Functional Requirement

1.4 References

- IEEE Std 830-1998
- Streamlit Documentation
- Eclipse Paho MQTT Documentation
- ESP32 Technical Documentation
- Python Machine Learning Libraries Documentation
- Research Papers on Smart Water Leakage Detection

2. OVERALL DESCRIPTION

2.1 Product Perspective

The system architecture consists of:

- Flow Sensor

- Pressure Sensor
- Moisture Sensor
- ESP32 Controller
- MQTT Broker
- AI/ML Prediction Engine
- Database
- Streamlit Dashboard
- Motor Relay Module
- Solenoid Valve
- Alarm Buzzer

Sensor readings are transmitted through MQTT to the AI engine. The AI model analyzes the readings and predicts potential leakages. Based on the analysis, the system performs automatic protection actions and updates the dashboard.

2.2 User Classes

User Role	Responsibility
Administrator	System configuration and monitoring
Maintenance Engineer	Leakage verification and repair
Operator	Dashboard monitoring and report analysis

2.3 Operating Environment

Hardware Requirements

- ESP32 Development Board
- Flow Sensor (YF-S201 or equivalent)
- Water Pressure Sensor
- Moisture Sensor
- Relay Module
- Solenoid Valve
- Alarm Buzzer
- Computer/Laptop

Software Requirements

- Python 3.10+

- Streamlit
- MQTT Broker (MQTTX)
- SQLite/MySQL
- VS Code
- Windows/Linux

2.4 Design Constraints

- Stable Wi-Fi connection required
- Sensor calibration required
- Real-time processing constraints
- MQTT broker availability

2.5 Assumptions

- Sensors are functioning correctly
- Continuous power supply is available
- MQTT server remains operational
- Users have access to the dashboard

3. SYSTEM FEATURES AND FUNCTIONAL REQUIREMENTS

3.1 Module 1: Sensor Data Acquisition

Req ID	Feature Description	Priority
FR-1.1	Read Flow Sensor Data	High
FR-1.2	Read Pressure Sensor Data	High
FR-1.3	Read Moisture Sensor Data	High
FR-1.4	Validate Sensor Values	High

3.2 Module 2: MQTT Communication

Req ID	Feature Description	Priority
FR-2.1	Publish sensor data to MQTT broker	High
FR-2.2	Subscribe to MQTT topics	High
FR-2.3	Handle communication failures	High
FR-2.4	Transmit control commands	High

3.3 Module 3: AI Leakage Detection

Req ID	Feature Description	Priority
FR-3.1	Data preprocessing	High
FR-3.2	Leakage prediction using ML model	High
FR-3.3	Leakage confidence score generation	High
FR-3.4	Leakage severity classification	High

3.4 Module 4: Alert and Monitoring

Req ID	Feature Description	Priority
FR-4.1	Generate leakage alerts	High
FR-4.2	Display AI confidence score	High
FR-4.3	Activate buzzer during leakage	High
FR-4.4	Store alert logs in database	High
FR-4.5	Dashboard notification generation	High

3.5 Module 5: Automatic Protection System

Req ID	Feature Description	Priority
FR-5.1	Turn motor OFF during leakage	High
FR-5.2	Close emergency valve	High
FR-5.3	Restore system when fault clears	Medium
FR-5.4	Maintain protection history logs	Medium

3.6 Module 6: Reporting and Visualization

Req ID	Feature Description	Priority
FR-6.1	Real-time dashboard updates	High
FR-6.2	Historical trend charts	Medium
FR-6.3	Leakage reports generation	Medium
FR-6.4	Data export functionality	Low

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interface

The Streamlit Dashboard shall provide:

- Live Sensor Monitoring
- Leakage Status Panel
- AI Confidence Indicator
- Real-Time Graphs
- Alert Notifications
- Historical Data Analysis
- System Control Panel

4.2 Hardware Interface

Hardware	Function
ESP32	Sensor Controller
Flow Sensor	Flow Monitoring
Pressure Sensor	Pressure Monitoring
Moisture Sensor	Surface Leakage Detection
Relay Module	Motor Control
Solenoid Valve	Water Supply Cutoff
Buzzer	Alarm Indication

4.3 Software Interface

Software	Function
Streamlit	Dashboard Development
MQTT Broker	Data Transmission
Python ML Model	Leakage Prediction
SQLite / MySQL	Data Storage
Paho MQTT	MQTT Client Communication

4.4 Communication Interface

Protocol

MQTT

Port

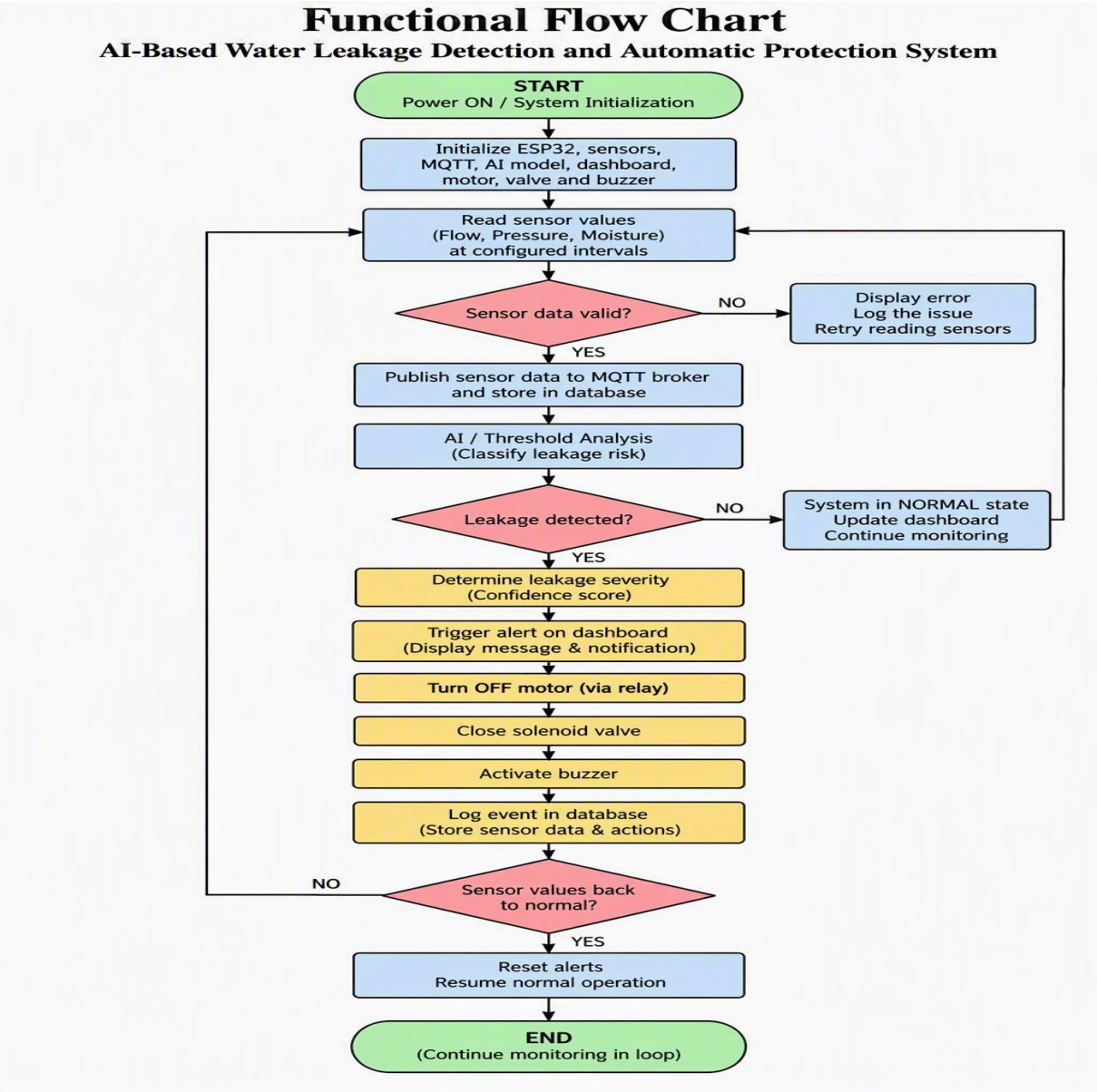
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5. NON-FUNCTIONAL REQUIREMENTS

Req ID	Requirement	Target
NFR-5.1	Detection Speed	Less than 3 Seconds
NFR-5.2	Availability	24×7 Operation
NFR-5.3	Reliability	≥ 98% Uptime
NFR-5.4	Security	Secure MQTT Authentication

NFR-5.5	Scalability	Multiple Sensor Nodes
NFR-5.6	Maintainability	Modular Architecture
NFR-5.7	Accuracy	AI Prediction Accuracy ≥ 90%

6. FUNCTIONAL FLOW CHART



7. TEST CASES

TC ID	Test Case	Input / Condition	Expected Result
TC-01	Normal Water Flow	Flow = 9–11 L/min, Pressure = 2.2–2.7 bar, Moisture = 0	System displays **NORMAL**
TC-02	Low Flow Detection	Flow < 7 L/min	**LEAKAGE** detected
TC-03	Low Pressure Detection	Pressure < 2.0 bar	**LEAKAGE** detected

TC-04	Moisture Detection	Moisture Sensor = WET (1)	**LEAKAGE** detected
TC-05	Multiple Abnormal Conditions	Low Flow + Low Pressure + Moisture Present	Leakage detected with high confidence
TC-06	Recovery After Leakage	Abnormal values return to normal range	Status changes back to **NORMAL**
TC-07	AI Confidence Prediction	Leakage condition supplied to ML model	AI confidence percentage displayed
TC-08	Motor Control	Leakage detected	Motor becomes **OFF**
TC-09	Valve Control	Leakage detected	Valve becomes **CLOSED**
TC-10	Buzzer Activation	NORMAL → LEAKAGE Transition	Buzzer generates warning alarm

Test Conditions for Leakage Classification

Condition	Flow Rate	Pressure	Moisture	Expected Output
Normal	9–11 L/min	2.2–2.7 bar	0	NORMAL
Minor Leakage	7–9 L/min	2.0–2.2 bar	0	Warning
Moderate Leakage	5–7 L/min	1.5–2.0 bar	1	Leakage
Critical Leakage	<5 L/min	<1.5 bar	1	Critical Leakage

8. FUTURE ENHANCEMENTS

- Mobile Application Support
- Cloud-Based Monitoring
- SMS/Email Notifications
- Smart City Water Network Integration
- Digital Twin Technology
- Advanced Deep Learning Models
- Predictive Maintenance Features

9. ACADEMIC REVIEW & SIGN-OFF

This Software Requirement Specification document has been submitted and reviewed by the academic supervisory panel :

Project Guide	Internal Examiner	HOD
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

*****END OF SOFTWARE REQUIREMENTS SPECIFICATION (SRS)*****